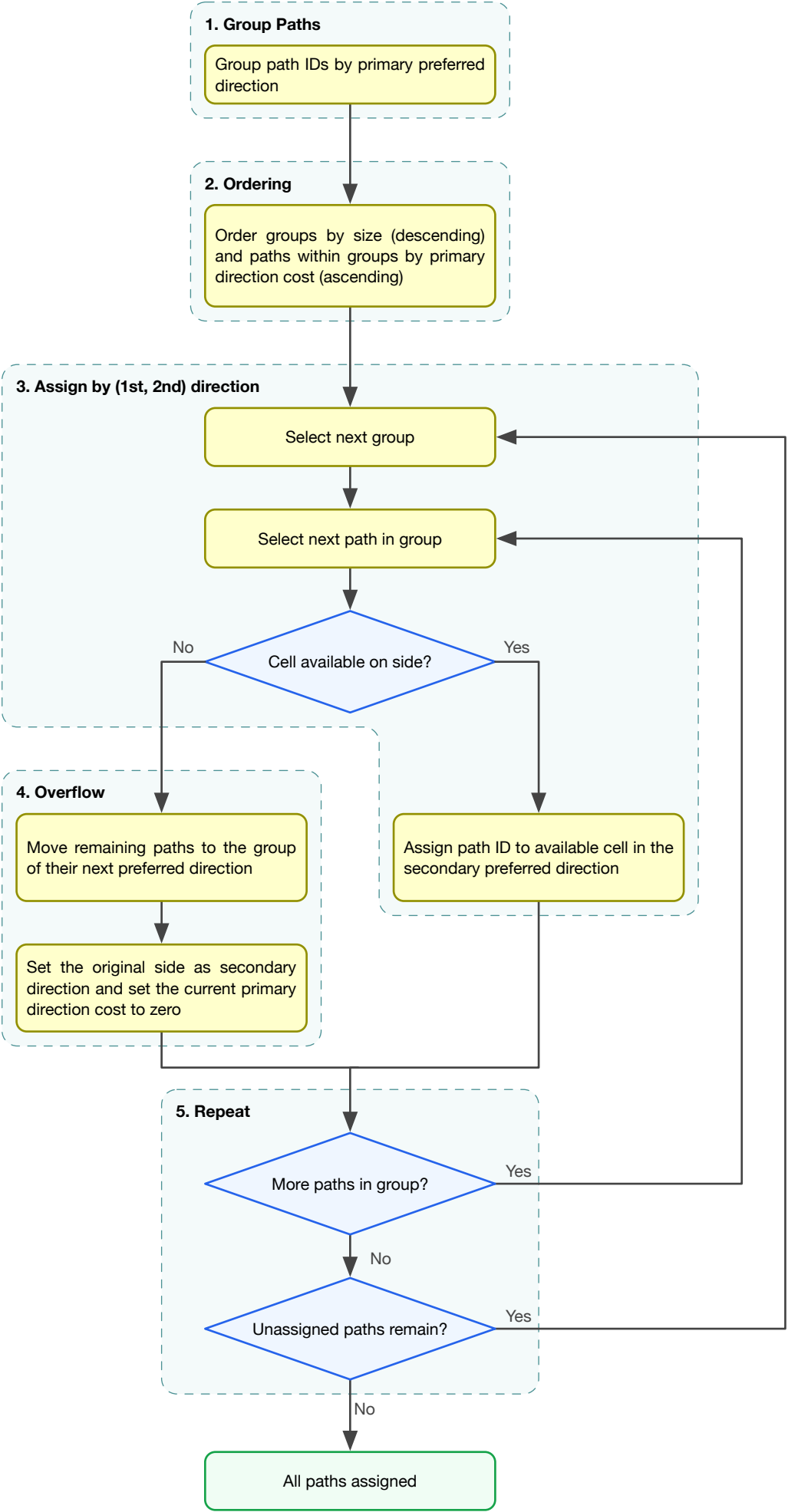
1. Group Paths
Group path IDs by primary preferred direction
2. Ordering
Order groups by size (descending) and paths within groups by primary direction cost (ascending)
3. Assign by (1st, 2nd) direction
Select next group
Select next path in group
Cell available on side?
No
Yes
4. Overflow
Move remaining paths to the group of their next preferred direction
Set the original side as secondary direction and set the current primary direction cost to zero
Assign path ID to available cell in the secondary preferred direction
5. Repeat
More paths in group?
Yes
No
Unassigned paths remain?
Yes
No
All paths assigned